



BOEDEKER PLASTICS, INC.
Plastics The Boedeker Way

1-800-444-3485

UHMW-PE Natural

COMPLIANCE INFO



(/Compliance/Prop-65)



(/Compliance#ROHS)



(/Compliance#REACH)



(/Compliance/Conflict-Minerals)



General Info

Ultra High Molecular Weight Polyethylene (UHMW PE) is a lightweight polymer (1/8 the weight of mild steel) that offers a low coefficient of friction, is self-lubricating, has high impact strength, is shatter-resistant, long-wearing, abrasion, and chemical and corrosion-resistant.

It meets FDA and USDA acceptance for food and pharmaceutical equipment. It performs well from cryogenic temperatures to 180 °F (82 °C).

It is one of the most common plastics used for sliding wear components on conveyors and automation equipment, bushings, bearings, and rollers in various food processing, pharmaceutical, and other industries.

Click for more information on the **[complete line of Polyethylene Plastics](http://boedeker.com/family/polyethylene)**
(<http://boedeker.com/family/polyethylene>)

Click for more information on the [complete line of UHMW Plastics \(/family/uhmw\)](#).

Key Properties

- Low coefficient of friction
- Abrasion/wear resistance
- High Impact strength
- Chemical and corrosion resistance
- Self-lubrication characteristics
- Zero moisture absorption
- Cryogenic capability
- Electrical resistivity

Limitations

- High Heat Environments
- High Loads
- Subject to Creep, Cold Flow
- Thermal Expansion
- Tight Tolerance Applications

Applications

- Bulk Material Handling – Hopper & Chute Lining Systems
- Bushings, Bearings & Rollers
- Conveyor Corner Tracks
- Conveyor Wear Strip
- Food & Beverage Processing Equipment Components
- Gears & Sprockets
- Linear Wear Pads
- Oil & Gas Subsea Components
- Wastewater Industry Components

Resin Trade Names

- GUR® UHMW-PE

Manufacturer Trade Names

- Duravar® UHMW Grades
- Lennite® UHMW Grades
- Polyslick® UHMW Grades
- Polystone® M UHMW Grades
- TIVAR® UHMW Grades

Boedeker Plastics, Inc. stocks & supplies plastic sheet, rod, tube, film and custom profiles and machine parts from these shapes that are produced from plastic resin. In many cases the manufacturer of the shape has their own product trade name. If your specification calls for a specific resin grade or manufacturer trade name, we can supply a shape made from that resin to meet your specification or suggest an equal. If you require a specific resin or brand it must be specified at the time of order.

Available Sizes

Available Colors

- Natural (White Opaque)
- Colors – May Require Minimums

Safety Data Sheet

Download Safety Data Sheet (<https://www.boedeker.com/Portals/0/Safety Data Sheets SDS/MCAM/MCAM-TIVAR-1000-UHMWPE-EN-SDS-01-191024.pdf?ver=S97kmoUnat7ZVolwr2ebow%3d%3d>)

Typical Properties of UHMW-PE Natural

DETAIL

Description	Value
Material Type	Semi-Crystalline Thermoplastic Polyethylene
Chemical Name	UHMW-PE Ultra High Molecular Weight Polyethylene
Additives	Unfilled
Color	Natural White

PHYSICAL

Property	Test	Unit of Measure	Value
Density	ASTM D792	g/cm ³	0.93
	ASTM D792	lb/in ³	0.034
Water Absorption, 24 hrs, Immersion	ASTM D570	% by wt.	<.01

MECHANICAL

Property	Test	Unit of Measure	Value
Tensile Strength	ASTM D638	psi	5,800
Tensile Modulus	ASTM D638	psi	80,000
Tensile Elongation at Break	ASTM D638	%	300
Flexural Strength	ASTM D790	psi	3,500
Flexural Modulus	ASTM D790	psi	88,000

DETAIL

Compressive Strength	ASTM D695	psi	3,000
Compressive Modulus	ASTM D695	psi	80,000
Hardness	ASTM D785	NA	Shore D
IZOD Impact-Notched	ASTM D256	ft-lb/in	No Break
Static Coefficient of Friction	ASTM D-1894	NA	.15 - .20
Dynamic Coefficient of Friction	ASTM D-1894	NA	.10 - .14

THERMAL

Property	Test	Unit of Measure	Value
Coefficient of Linear Thermal Expansion	ASTM D696	x 10-5 in./in./°F	11
Heat Deflection Temp	ASTM D648	°C at 264 psi	46
	ASTM D648	°F at 264 psi	116
Melting Point Temp	ASTM D3418	°C	136
	ASTM D3418	°F	275
Max Continuous Operating Temp		°C	82
		°F	180
Minimum Operating Temp		°C	-200
		°F	-328

DETAIL

Thermal Conductivity	ASTM C117	BTU-in/ft ² -hr-°F	2.84
	ASTM C117	x 10 ⁴ cal/cm-sec-°C	10
Flammability Rating	UL94	NA	HB

ELECTRICAL

Property	Test	Unit of Measure	Value
Dielectric Strength	ASTM D149	V/mil	1,150
Dielectric Constant at 1 MHz	ASTM D150	NA	2.3
Dissipation Factor at 1 MHz	ASTM D150	NA	0.001
Surface Resistivity	ASTM D257	ohm/square	> 10 ¹⁵

COMPLIANCE

Description	Value
FDA	Yes
USDA	Yes
NSF	No
3-A Dairy	Yes
Canadian Food Inspection Agency	Yes
USP Class VI	No

OTHER

Description	Value
UV Resistant	No

DETAIL

Relative Cost	\$
Machinability	1

Relative cost based on a scale of one dollar sign (\$) as the lowest to ten dollar signs (\$\$\$\$\$\$\$\$) as the highest.

Machinability based on a scale of 1 to 10, with 1 being the easiest and 10 being the most difficult to machine. NOTE: All information contained on www.boedeker.com, downloadable documents and printed literature is intended for technical reference only and considered to be accurate for reference purposes only. Boedeker Plastics, Inc. makes no guarantee and offers no warranty for fitness in use and strongly recommends the user validate any plastic, machined part, or product in a specific application for fitness in use.